Лабораторная работа 6

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1. Вывести список всех процессов системы.

```
ubuntu@ubuntu:~$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.8  0.4  23388  8592 ?        Ss   12:00   0:06 /sbin/init au
root           2  0.0  0.0      0     0 ?        S    12:00   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    12:00   0:00 [pool_workque
root           4  0.0  0.0      0     0 ?        I<   12:00   0:00 [kworker/R-rc
root           5  0.0  0.0      0     0 ?        I<   12:00   0:00 [kworker/R-sy
root           6  0.0  0.0      0     0 ?        I<   12:00   0:00 [kworker/R-kv
root           7  0.0  0.0      0     0 ?        I<   12:00   0:00 [kworker/R-sl
root           8  0.0  0.0      0     0 ?        I<   12:00   0:00 [kworker/R-ne
root          10  4.5  0.0      0     0 ?        I    12:00   0:34 [kworker/0:1-
root          11  0.0  0.0      0     0 ?        I<   12:00   0:00 [kworker/0:0H
root          12  1.5  0.0      0     0 ?        I    12:00   0:11 [kworker/u4:0
root          13  0.0  0.0      0     0 ?        I<   12:00   0:00 [kworker/R-mm
root          14  0.0  0.0      0     0 ?        I    12:00   0:00 [rcu_tasks_kt
root          15  0.0  0.0      0     0 ?        I    12:00   0:00 [rcu_tasks_ru
root          16  0.0  0.0      0     0 ?        I    12:00   0:00 [rcu_tasks_tr
root          17  1.8  0.0      0     0 ?        S    12:00   0:13 [ksoftirqd/0]
root          18  0.4  0.0      0     0 ?        I    12:00   0:03 [rcu_preempt]
root          19  0.0  0.0      0     0 ?        S    12:00   0:00 [rcu_exp_par_
root          20  0.0  0.0      0     0 ?        S    12:00   0:00 [rcu_exp_gp_k
root          21  0.0  0.0      0     0 ?        S    12:00   0:00 [migration/0]
root          22  0.0  0.0      0     0 ?        S    12:00   0:00 [idle_inject/
root          23  0.0  0.0      0     0 ?        S    12:00   0:00 [cpuhp/0]
```

2. Вывести дерево процессов.

```
ubuntu      6298  0.0  0.0    452   136 ?        D    12:13   0:00 /snap/snap-st
ubuntu@ubuntu:~$ pstree
systemd─┬─ModemManager───3*[{ModemManager}]
        ├─NetworkManager───3*[{NetworkManager}]
        ├─accounts-daemon───3*[{accounts-daemon}]
        ├─avahi-daemon───avahi-daemon
        ├─casper-md5check
        ├─colord───3*[{colord}]
        ├─cron
        ├─cups-browsed───3*[{cups-browsed}]
        ├─cupsd
        ├─dbus-daemon
        ├─gdm3─┬─gdm-session-wor─┬─gdm-x-session─┬─Xorg───{Xorg}
        │      │                 │               ├─gnome-session-b───3*[{gnome-session-b}]
        │      │                 │               └─3*[{gdm-x-session}]
        │      │                 └─3*[{gdm-session-wor}]
        │      └─3*[{gdm3}]
        ├─gnome-remote-de───3*[{gnome-remote-de}]
        ├─2*[kerneloops]
        ├─polkitd───3*[{polkitd}]
        ├─power-profiles-───3*[{power-profiles-}]
        ├─python3.10───python3.10───rsync───rsync───rsync
        ├─rsyslogd───3*[{rsyslogd}]
        ├─rtkit-daemon───2*[{rtkit-daemon}]
```

3. С помощью команды top получить список 5 процессов, потребляющих наибольшее количество процессорного времени.

Командой top вывожу динамически обновляющийся список всех процессов

```
ubuntu@ubuntu:~$ top

top - 12:32:12 up 31 min,  1 user,  load average: 3.70, 4.34, 5.33
Tasks: 217 total,   2 running, 215 sleeping,   0 stopped,   0 zombie
%Cpu(s):  3.7 us,  7.0 sy,  0.9 ni,  0.0 id, 74.3 wa,  0.0 hi, 14.0 si,  0.0 st
MiB Mem :  1968.7 total,     86.4 free,   1886.4 used,    775.7 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.     82.4 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   6897 root      20   0  342816 129352  13156 D  26.5   6.4   0:03.74 appstreamcli
   2516 ubuntu    20   0 3454040 239556  23780 S   3.0  11.9   1:27.75 gnome-shell
   4118 root      39  19    3776    904    648 D   2.3   0.0   0:42.34 casper-md5check
     47 root      20   0       0      0      0 S   1.7   0.0   0:15.37 kswapd0
   6387 root      20   0       0      0      0 D   1.3   0.0   0:00.15 kworker/u4:3+flush-+
   6418 root       0 -20       0      0      0 I   1.3   0.0   0:05.78 kworker/0:0H-kblockd
   6424 root      20   0       0      0      0 I   1.3   0.0   0:02.63 kworker/0:0-ata_sff
     12 root      20   0       0      0      0 I   0.3   0.0   0:22.27 kworker/u4:0-ext4-r+
     17 root      20   0       0      0      0 S   0.3   0.0   0:32.26 ksoftirqd/0
   1278 root      39  19       0      0      0 S   0.3   0.0   0:00.19 arc_reap
   1315 root      20   0       0      0      0 I   0.3   0.0   0:18.37 kworker/u4:8-events+
```

Нажимаю f для сортировки, после чего выбирают time+, затем q для выхода из меню. Теперь список отсортирован по потреблению времени.

```
Fields Management for window 1:Def, whose current sort field is TIME+
   Navigate with Up/Dn, Right selects for move then <Enter> or Left commits,
   'd' or <Space> toggles display, 's' sets sort.  Use 'q' or <Esc> to end!

* PID      = Process Id  PGRP     = Process Gr  OOMs     = OOMEM Scor  RSS      = Res Mem (s
* USER     = Effective   TTY      = Controllin  ENVIRON  = Environmen  PSS      = Proportion
* PR       = Priority    TPGID    = Tty Proces  vMj      = Major Faul  PSan     = Proportion
* NI       = Nice Value  SID      = Session Id  vMn      = Minor Faul  PSfd     = Proportion
* VIRT     = Virtual Im  nTH      = Number of   USED     = Res+Swap S  PSsh     = Proportion
* RES      = Resident S  P        = Last Used   nsIPC    = IPC namesp  USS      = Unique RSS
* SHR      = Shared Mem  TIME     = CPU Time    nsMNT    = MNT namesp  ioR      = I/O Bytes
* S        = Process St  SWAP     = Swapped Si  nsNET    = NET namesp  ioRop    = I/O Read O
* %CPU     = CPU Usage   CODE     = Code Size   nsPID    = PID namesp  ioW      = I/O Bytes
* %MEM     = Memory Usa  DATA     = Data+Stack  nsUSER   = USER names  ioWop    = I/O Write
* TIME+    = CPU Time,   nMaj     = Major Page  nsUTS    = UTS namesp  AGID     = Autogroup
* COMMAND  = Command Na  nMin     = Minor Page  LXC      = LXC contai  AGNI     = Autogroup
  PPID     = Parent Pro  nDRT     = Dirty Page  RSan     = RES Anonym  STARTED  = Start Time
  UID      = Effective   WCHAN    = Sleeping i  RSfd     = RES File-b  ELAPSED  = Elapsed Ru
```

```
top - 12:41:36 up 40 min,  1 user,  load average: 1.70, 2.22, 3.84
Tasks: 213 total,   1 running, 212 sleeping,   0 stopped,   0 zombie
%Cpu(s):  1.2 us, 22.1 sy,  0.0 ni,  0.0 id, 66.0 wa,  0.0 hi, 10.7 si,  0.0 st
MiB Mem :  1968.7 total,     91.5 free,   1823.2 used,    845.2 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.    145.5 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0   23228   8348   3484 S   0.0   0.4   6:36.06 systemd
   2516 ubuntu    20   0 3454040 242892  27256 S   1.0  12.0   1:53.69 gnome-shell
   2145 ubuntu    20   0   21568   5336   2008 S   0.0   0.3   1:42.97 systemd
   6625 root      20   0   61072  24076   1592 S   0.0   1.2   1:13.18 python3.10
   1654 root      20   0 1847964  15000    420 S   0.0   0.7   1:08.12 snapd
     17 root      20   0       0      0      0 S   0.0   0.0   0:39.46 ksoftirqd/0
   2293 ubuntu    20   0  318232  52252  13360 S   1.3   2.6   0:37.35 Xorg
   4094 root      20   0  646852  62156   3908 S   0.0   3.1   0:26.86 python3.10
     12 root      20   0       0      0      0 I   0.3   0.0   0:24.34 kworker/u4:0-ext4-r+
   6393 root      20   0       0      0      0 I   0.0   0.0   0:23.90 kworker/0:2-cgroup_+
   1315 root      20   0       0      0      0 I   0.0   0.0   0:20.35 kworker/u4:8-events+
     47 root      20   0       0      0      0 S   0.3   0.0   0:16.97 kswapd0
   7733 root      20   0   43696  28508   2756 S  19.3   1.4   0:14.58 dpkg
   6424 root      20   0       0      0      0 I   0.0   0.0   0:14.51 kworker/0:0-cgroup_+
   6151 ubuntu    20   0  776548  18744   7560 S   0.3   0.9   0:13.72 gnome-terminal-
```

После нажимаю n и ввожу число 5 чтобы оставить только 5 верхних строчек.

```
top - 12:42:18 up 41 min,  1 user,  load average: 1.41, 2.08, 3.72
Tasks: 213 total,   2 running, 211 sleeping,   0 stopped,   0 zombie
%Cpu(s):  1.2 us, 30.5 sy,  0.0 ni,  0.0 id, 56.0 wa,  0.0 hi, 12.4 si,  0.0 st
MiB Mem :   1968.7 total,     83.0 free,   1826.1 used,    851.8 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.    142.6 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   2516 ubuntu    20   0 3454040 243352  27716 S   4.3  12.1   1:42.07 gnome-shell
     17 root      20   0       0      0      0 S   0.0   0.0   0:39.55 ksoftirqd/0
   2293 ubuntu    20   0  318232  52896  14004 S   4.3   2.6   0:37.97 Xorg
     12 root      20   0       0      0      0 I   1.0   0.0   0:24.68 kworker/u4:0-writeb+
   1654 root      20   0 1847964  18392   3812 S   0.0   0.9   0:24.13 snapd
```

4. Найти 2 процесса, имеющих более двух потоков. Использовать состояние процесса.

   Ввожу команду top, после чего нажимаю o для фильтра и ввожу S=I, чтобы отображались только процессы с двумя и более потоками.

```
ubuntu@ubuntu: ~
top - 12:55:55 up 55 min,  1 user,  load average: 1.72, 1.69, 2.64
Tasks: 218 total,   3 running, 212 sleeping,   0 stopped,   3 zombie
%Cpu(s):  3.7 us, 34.6 sy,  0.0 ni,  0.0 id,  0.7 wa,  0.0 hi, 61.0 si,  0.0 st
MiB Mem :   1968.7 total,     77.2 free,   1897.7 used,    771.9 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.     71.0 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   7732 root      20   0       0      0      0 I   3.4   0.0   0:10.66 kworker/0:1-ata_sff
     18 root      20   0       0      0      0 I   0.6   0.0   0:07.00 rcu_preempt
  13784 root      20   0       0      0      0 I   0.3   0.0   0:00.43 kworker/u4:3-events+
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rc+
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flus+
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
     12 root      20   0       0      0      0 I   0.0   0.0   0:25.36 kworker/u4:0-loop2
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu+
     14 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
     15 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthr+
     16 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kth+
     25 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-inet_frag+
     31 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-writeback
```

```
top - 12:58:49 up 58 min,  1 user,  load average: 11.89, 5.42, 3.82
Tasks: 223 total,   2 running, 219 sleeping,   0 stopped,   2 zombie
%Cpu(s):  3.2 us, 44.6 sy,  0.0 ni,  0.0 id, 36.9 wa,  0.0 hi, 15.3 si,  0.0 st
MiB Mem :   1968.7 total,     77.6 free,   1883.5 used,    804.9 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.     85.2 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
```

5. Используя команду top, изменить приоритеты 2 процессов.

После вызова команды sudo top, чтобы вызвать команду в режиме суперпользователя, нажимаю r и ввожу pid процесса.

```
top - 12:59:49 up 59 min,  1 user,  load average: 6.30, 4.96, 3.76
Tasks: 225 total,   1 running, 222 sleeping,   0 stopped,   2 zombie
%Cpu(s):  0.0 us, 29.4 sy,  5.9 ni,  0.0 id, 47.1 wa,  0.0 hi, 17.6 si,  0.0 st
MiB Mem :   1968.7 total,     75.8 free,   1908.8 used,    759.3 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.     59.9 avail Mem
PID to renice [default pid = 14928] 8
    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
  14928 root      39  19  210776  70472  29184 D   4.8   3.5   0:03.45 apt-check
      1 root      20   0   23484   5940   1076 S   0.0   0.3   0:16.30 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_rele+
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rc+
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flus+
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
```

Выбираем процесс 1 и меняем ему приоритет на -10

```
MiB Mem :   1968.7 total,     74.5 free,    978.1 used,   1101.3 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.    990.6 avail Mem
Renice PID 1 to value -10
    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   1951 vboxuser  20   0 3462708 382340 144156 S  13.0  19.0   0:21.00 gnome-s+
   3916 vboxuser  20   0  554060  53088  42316 S   4.0   2.6   0:00.82 gnome-t+
   3785 root      20   0       0      0      0 I   0.3   0.0   0:00.17 kworker+
```

В столбцах pr и ni можно увидеть изменения.

```
   1951 vboxuser  20   0 3454572 320716  82700 S   0.7  15.9   1:05.59 gnome-s+
    284 root      20   0       0      0      0 I   0.3   0.0   0:03.72 kworker+
   3916 vboxuser  20   0  624392  40424  28500 S   0.3   2.0   0:03.43 gnome-t+
   4179 root      20   0   14536   5828   3652 R   0.3   0.3   0:00.32 top
      1 root      10 -10   23272  14076   9476 S   0.0   0.7   0:33.07 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_wo+
```

Проделываю то же самое с процессом 2.

```
MiB Mem :   1968.7 total,    526.9 free,    993.8 used,    632.6 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.    974.9 avail Mem
Renice PID 2 to value -10
    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   1951 vboxuser  20   0 3454572 320716  82700 S   0.7  15.9   1:06.52 gnome-s+
    284 root      20   0       0      0      0 I   0.3   0.0   0:03.90 kworker+
    835 root      20   0 1865340  39624  24080 S   0.3   2.0   0:46.98 snapd
   3899 root      20   0       0      0      0 I   0.3   0.0   0:01.25 kworker+
   4179 root      20   0   14536   5828   3652 R   0.3   0.3   0:00.49 top
      1 root      10 -10   23272  14076   9476 S   0.0   0.7   0:33.07 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_wo+
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
```

```
   4179 root      20   0   14536   5828   3652 R   0.5   0.3   0:00.51 top
      1 root      10 -10   23272  14076   9476 S   0.0   0.7   0:04.12 systemd
      2 root      10 -10       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_wo+
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
```

6. Получить список открытых файлов пользователя

```
vboxuser@ubuntu:~$ lsof -u vboxuser
```

```
/lib/x86_64-linux-gnu/libsharpyuv.so.0.0.1
gnome-she 1951 vboxuser  mem       REG          8,2     14344 1322820 /usr
/lib/x86_64-linux-gnu/libwayland-egl.so.1.22.0
gnome-she 1951 vboxuser  mem       REG          8,2    510592 1322808 /usr
/lib/x86_64-linux-gnu/libvulkan.so.1.3.275
gnome-she 1951 vboxuser  mem       REG          8,2   1189616 1322305 /usr
/lib/x86_64-linux-gnu/libepoxy.so.0.0.0
gnome-she 1951 vboxuser  mem       REG          8,2     22600 1322498 /usr
/lib/x86_64-linux-gnu/libkeyutils.so.1.10
gnome-she 1951 vboxuser  mem       REG          8,2     18504 1322241 /usr
/lib/x86_64-linux-gnu/libcom_err.so.2.1
gnome-she 1951 vboxuser  mem       REG          8,2     41804 1323961 /usr
/lib/x86_64-linux-gnu/girepository-1.0/Gsk-4.0.typelib
gnome-she 1951 vboxuser  mem       REG          8,2    570548 1323968 /usr
/lib/x86_64-linux-gnu/girepository-1.0/Gtk-4.0.typelib
gnome-she 1951 vboxuser  mem       REG          8,2     14972 1323975 /usr
/lib/x86_64-linux-gnu/girepository-1.0/NMA4-1.0.typelib
gnome-she 1951 vboxuser  mem       REG          8,2     19176 1313326 /usr
/lib/gnome-shell/Gvc-1.0.typelib
gnome-she 1951 vboxuser  mem       REG          8,2    340688 1323974 /usr
/lib/x86_64-linux-gnu/girepository-1.0/NM-1.0.typelib
gnome-she 1951 vboxuser  mem       REG          8,2     45580 1313328 /usr
/lib/gnome-shell/St-14.typelib
```

7. Получить текущее состояние системной памяти

```
vboxuser@ubuntu:~$ free
               total        used        free      shared  buff/cache   available
Mem:         2015944     1024332      485340       35696      696216      991612
Swap:              0           0           0
```

8. Получить справку об использовании дискового пространства.

```
vboxuser@ubuntu:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           197M  1.7M  196M   1% /run
/dev/sda2        25G  5.6G   18G  24% /
tmpfs           985M     0  985M   0% /dev/shm
tmpfs           5.0M  8.0K  5.0M   1% /run/lock
tmpfs           197M  156K  197M   1% /run/user/1000
/dev/sr0        6.0G  6.0G     0 100% /media/vboxuser/Ubuntu 24.04.3 LTS amd64
```

9. Вывести информацию о каком-либо процессе, используя содержимое каталога /proc

```
vboxuser@ubuntu:~$ ls /proc/1
ls: cannot read symbolic link '/proc/1/cwd': Permission denied
ls: cannot read symbolic link '/proc/1/root': Permission denied
ls: cannot read symbolic link '/proc/1/exe': Permission denied
arch_status          fdinfo              ns              smaps_rollup
attr                 gid_map             numa_maps       stack
autogroup            io                  oom_adj         stat
auxv                 ksm_merging_pages   oom_score       statm
cgroup               ksm_stat            oom_score_adj   status
clear_refs           latency             pagemap         syscall
cmdline              limits              patch_state     task
comm                 loginuid            personality     timens_offsets
coredump_filter      map_files           projid_map      timers
cpu_resctrl_groups   maps                root            timerslack_ns
cpuset               mem                 sched           uid_map
cwd                  mountinfo           schedstat       wchan
environ              mounts              sessionid
exe                  mountstats          setgroups
fd                   net                 smaps
vboxuser@ubuntu:~$
```

10. Вывести информацию о процессоре ПК, используя содержимое каталога /proc

```
vboxuser@ubuntu:~$ cat /proc/cpuinfo
processor       : 0
vendor_id       : GenuineIntel
cpu family      : 6
model           : 78
model name      : Intel(R) Core(TM) i7-6500U CPU @ 2.50GHz
stepping        : 3
microcode       : 0xffffffff
cpu MHz         : 2591.998
cache size      : 4096 KB
physical id     : 0
siblings        : 1
core id         : 0
cpu cores       : 1
apicid          : 0
initial apicid  : 0
fpu             : yes
fpu_exception   : yes
cpuid level     : 22
wp              : yes
flags           : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov
pat pse36 clflush mmx fxsr sse sse2 ht syscall nx rdtscp lm constant_tsc rep_goo
d nopl xtopology nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 p
cid sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm abm
3dnowprefetch pti fsgsbase bmi1 avx2 bmi2 invpcid rdseed adx clflushopt arat md_
clear flush_l1d arch_capabilities
bugs            : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds
swapgs itlb_multihit srbds mmio_stale_data retbleed gds bhi its
bogomips        : 5183.99
clflush size    : 64
```

11. Вывести список модулей, используемых в настоящий момент ядром ОС.

```
vboxuser@ubuntu:~$ lsmod
Module                  Size  Used by
isofs                  61440  1
snd_seq_dummy          12288  0
snd_hrtimer            12288  1
qrtr                   53248  2
intel_rapl_msr         20480  0
intel_rapl_common      53248  1 intel_rapl_msr
intel_pmc_core        122880  0
pmt_telemetry          16384  1 intel_pmc_core
pmt_class              16384  1 pmt_telemetry
intel_vsec             20480  1 intel_pmc_core
snd_intel8x0           53248  1
snd_ac97_codec        196608  1 snd_intel8x0
ac97_bus               12288  1 snd_ac97_codec
snd_pcm               192512  2 snd_intel8x0,snd_ac97_codec
snd_seq_midi           24576  0
snd_seq_midi_event     16384  1 snd_seq_midi
polyval_clmulni        12288  0
snd_rawmidi            57344  1 snd_seq_midi
polyval_generic        12288  1 polyval_clmulni
ghash_clmulni_intel    16384  0
snd_seq               122880  9 snd_seq_midi,snd_seq_midi_event,snd_seq_dummy
sha256_ssse3           32768  0
sha1_ssse3             32768  0
aesni_intel           122880  0
snd_seq_device         16384  3 snd_seq,snd_seq_midi,snd_rawmidi
snd_timer              53248  3 snd_seq,snd_hrtimer,snd_pcm
joydev                 32768  0
crypto_simd            16384  1 aesni_intel
```